

Stress-Testing Product-Gated Clustering and Bounded Priority Ranking for Flood-Rescue Reports: A Synthetic Study

Thanh-Trong Nguyen¹[0009-0007-6976-5131], Ngoc-Anh Le¹[0009-0006-7453-1009],
Nhu-Quynh Nguyen¹[0009-0008-7396-697X], Tuong-Hung
Cao¹[0009-0003-5886-2666], Hung-Thinh Ngo¹[0009-0004-8668-4462], Xuan-Phuong
Chau¹, Lan Phuong Phan¹, and Thanh-Khoa Nguyen¹[0000-0002-0012-1158]*

College of Information and Communication Technology, Can Tho University, Can
Tho, Vietnam

trongb2305615@student.ctu.edu.vn
{anhb2303861, quynhb2303777, hungb2303873,
thinhb2303904}@student.ctu.edu.vn
{cxphuong, pplan, ntkhoa}@ctu.edu.vn

Abstract. Operational flood streams contain duplicate, incomplete, and adversarial reports, while clustering and priority scores are often evaluated without measuring their scheduling consequences. We audit a fail-closed batch pipeline in which observable reports form a sparse graph, unresolved reports go to review, duplicate families feed a bounded score, and evaluator-only truth is joined after predicted-cluster scheduling. The clustering benchmark contains 80 synthetic runs from generator 3.0.0, of which 40 are held out for testing. Additive Louvain has mean ARI 0.9191 and Product Louvain .9072. Their mean paired difference is -0.0118 (unadjusted bootstrap 95% CI $[-0.0226, -0.0024]$), while the Holm-adjusted Wilcoxon test is not significant ($p = .2560$); no composition advantage is established. Under fixed configurations, location and timestamp noise reduce accuracy, and fivefold exact transport copies reduce all four graph variants to mean ARI near .30. In a separate candidate-4.1 bundle, the revised score has NDCG@5 .6669, a lower point estimate than the random comparator’s .6763, and shows no general alignment advantage even under oracle incident grouping. Its family-wise value is invariant to exact copies when grouping and inputs are fixed, but coordinated high-confidence campaigns remain a failure mode. Dispatch shows no general benefit over the legacy score and worse harm and deadline outcomes than nearest-first. These within-generator synthetic diagnostics delineate mathematical safeguards without establishing source authenticity, policy validity, field benefit, or deployment readiness.

Keywords: Flood rescue · Graph clustering · Synthetic stress test · Priority ranking · Negative results.

* Corresponding author.

1 Introduction and Related Work

During emergencies, crisis streams mix repeated reports of one event with nearby but distinct events, missing fields, and nonincident noise. Social-sensor and emergency-message research addresses volume, verification, and event extraction [11]. CrisisMMD and FloodNet capture useful disaster modalities, although FloodNet is an aerial post-flood scene-understanding dataset rather than a multimodal report stream [1,10]. These resources do not jointly provide physical-incident linkage, rescue priority, and field dispatch outcomes.

Social sensing and event extraction. Social-media users can act as real-time sensors [11], but the usual endpoint is event detection or classification rather than the consequences of sending a field resource to a predicted incident.

Consolidation and clustering. Graph community detection with Louvain or Leiden [3,15], and density methods such as DBSCAN and HDBSCAN [5] offer plausible consolidation mechanisms. Multiplicative spatial-range terms occur in bilateral filtering [14]; we adapt that structure to a product graph, while testing the exact gate and its bounds here. Throughout the paper, *Product Louvain* denotes Louvain applied to a graph whose temporal and contextual affinity is multiplied by a spatial gate; *Product Leiden* keeps that graph and changes only the community detector. *Additive Louvain* instead sums the spatial, temporal, and contextual terms. The *matched-density Additive* diagnostic adjusts its sparsification settings without labels to approximate the Product graph’s retained-edge fraction and mean degree.

Humanitarian logistics and dispatch. Relief-routing models expose competing equity, efficacy, and efficiency objectives [6]. We use this perspective to motivate a bounded priority ranking, but boundedness is not an expert-endorsed rescue policy. A key gap is propagation of false splits, merges, and noise into a scheduler that cannot consult incident truth.

We address these gaps with a fail-closed batch pipeline: reports with observable location and time enter a sparse graph, incomplete reports go to manual review, duplicate families feed a bounded ranking, and evaluator-only truth joins after predicted-cluster scheduling. We ask: (RQ1) does independently selected product composition improve clustering over additive composition? (RQ2) does the duplicate-aware score align with an independently generated benefit under report-level stress tests? (RQ3) how do predicted split, merge, and noise errors propagate into dispatch? RQ1 and RQ2/RQ3 use separate artifact bundles; their results characterize different pipeline stages rather than one empirical chain over a common sample.

The contributions delineate safeguards at three stages. At clustering, the spatial gate supplies a checkable conditional edge-localization bound. At ranking, family-wise aggregation supplies an exact-copy invariance property when evidence grouping and score inputs are fixed. At dispatch, a paired audit tests

whether these upstream properties translate to simulated outcomes and exposes cases where they do not. The pipeline also makes the manual-review and evaluator-only boundaries explicit. These are diagnostic and implementation checks, not evidence of general superiority or field utility; the artifact is not a validated rescue policy.

2 Observable Pipeline and Methods

Figure 1 fixes the information boundary. Receipt-visible fields flow through graph construction, ranking, and a batch scheduler. Missing location or event time routes a report to review. Incident identity, benefit, deadline, and harm are stored separately and join only after a schedule exists.

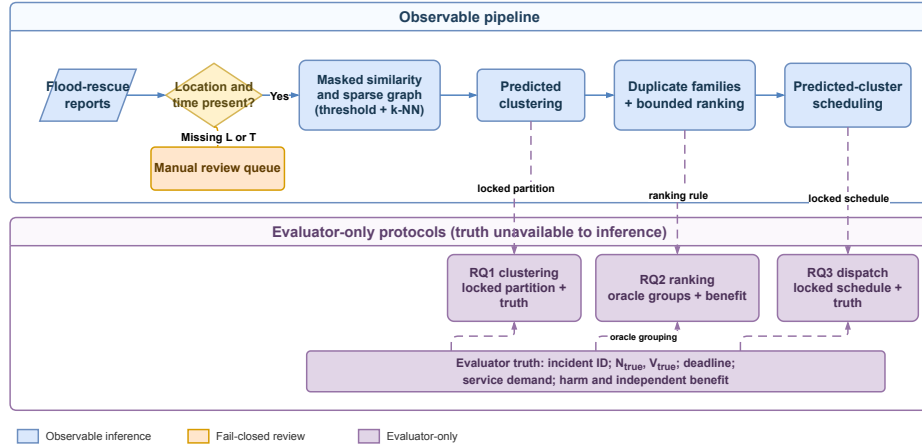


Fig. 1. Observable pipeline (solid), manual-review path (orange), and evaluator-only truth join (dashed). The scheduler receives predicted clusters, never incident truth.

Algorithms 1 and 2 restate admission, graph construction, duplicate-aware scoring, and scheduling; neither receives the evaluator-only truth join.

2.1 Report Attributes and Missingness

Each report is represented by the compact tuple

$$r_i = (L_i, T_i, F_i, E_i, N_i, V_i, Q_i),$$

where L_i is a GPS location and T_i is the observable report timestamp (“created_at”). Flood evidence F_i and urgency evidence E_i are in $[0, 1]$; $N_i \in \mathbb{Z}_{\geq 0}$ is the reported trapped-person count; $V_i \geq 0$ is a vulnerability-evidence value; and

$Q_i \in [0, 1]$ is a provenance/confidence heuristic. Q_i is not a calibrated probability of truth. The implementation also carries observable support fields such as image presence, source type, province, and missing-field flags; these support Q_i and duplicate fingerprints but are not additional latent labels.

The score Q_i is computed from observable image presence and the number of nearby corroborating observable payloads (counted by unique payload identity) using the proposed heuristic

$$Q_i = \text{sigmoid}(b_0 + b_1 \mathbf{1}\{\text{image}_i\} + b_2 \log(1 + n_i^{\text{corrob}})), \quad (1)$$

$$(b_0, b_1, b_2) = (-.2, 1.4, .9).$$

Here $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$, $\mathbf{1}\{\text{image}_i\}$ is one exactly when report i contains an image, and $n_i^{\text{corrob}} \in \mathbb{Z}_{\geq 0}$ counts distinct nearby observable payloads. The dimensionless b_0, b_1, b_2 are the fixed intercept, image, and corroboration coefficients; consequently $Q_i \in (0, 1)$. This is a pipeline rule, not a calibrated truth model. No independent receipt timestamp or source family is assumed.

Following the general missing-variable kernel problem [12], we use the following fail-closed construction. The admission indicator $a_i = \mathbf{1}\{L_i, T_i \text{ are observed}\}$. Thus $a_i a_j = 0$ prevents an edge involving a report whose geographic or temporal separation is undefined; such a report follows the manual-review path. Missing F_i or E_i is not zero-imputed: let $I_F(i, j), I_E(i, j) \in \{0, 1\}$ indicate that both reports contain the corresponding observation and let $m_{ij} = I_F(i, j) + I_E(i, j)$; then

$$C_{ij} = \frac{m_{ij}}{2} \exp\left[-I_F(i, j) \frac{|F_i - F_j|}{\tau_F} - I_E(i, j) \frac{|E_i - E_j|}{\tau_E}\right], \quad C_{ij} = 0 \text{ if } m_{ij} = 0. \quad (2)$$

$\tau_F, \tau_E > 0$ are dimensionless flood- and urgency-difference scales, so $C_{ij} \in [0, 1]$. When $m_{ij} = 2$, both contextual fields are used; when $m_{ij} = 1$, the maximum contextual similarity is discounted to $1/2$; and when $m_{ij} = 0$, no contextual agreement is inferred. In the current 80-run artifact, every report has location, event time, flood, and urgency. Hence $a_i = 1$ and $m_{ij} = 2$ throughout the evaluated data, so Eq. 2 uses its full two-field exponential branch. The missingness extension is specified and unit-tested, but missingness robustness is not claimed as an empirical result of this benchmark.

2.2 Similarity Graph and Derived Bounds

For Haversine distance d_{ij} and time difference Δt_{ij} , define the Gaussian affinity [8] and exponential temporal decay [18]:

$$G_{ij} = \exp[-d_{ij}^2/(2\sigma^2)], \quad T_{ij} = \exp[-\Delta t_{ij}/\tau_t],$$

$$w_{ij}^\times = a_i a_j G_{ij} (\beta T_{ij} + \gamma C_{ij}), \quad w_{ij}^+ = a_i a_j (\alpha G_{ij} + \beta T_{ij} + \gamma C_{ij}). \quad (3)$$

d_{ij} is measured in metres and Δt_{ij} in minutes; $\sigma > 0$ and $\tau_t > 0$ are the corresponding spatial and temporal bandwidths. The dimensionless $G_{ij}, T_{ij}, C_{ij} \in [0, 1]$ enter the product and additive weights $w_{ij}^\times$ and w_{ij}^+ with nonnegative α, β, γ .

Superscripts $\times$ and $+$ identify composition, not multiplication or addition over report indices. The exponential form is a precedent, not a claim that the complete missingness-aware expression is copied from prior work [2]. The product is a spatial gate motivated by spatial-range products such as bilateral filtering [14]; both compositions and all weights here are defined by this paper. The implementation forms the eligible pair matrix, computes an empirical nonzero-weight quantile, removes weights at or below that threshold, retains the endpoint-wise top- k among the remaining directed neighbours, and symmetrizes their union before community detection. Thus an undirected node may have degree above k when another node selects it. Product and additive candidates are selected independently; the estimand is therefore a pipeline comparison, not an operator-only causal effect. We treat this independently calibrated contrast as the primary estimand for RQ1 because it asks whether each tuned pipeline provides a net empirical benefit. Section 4 also reports a matched-density contrast as a secondary diagnostic. It reduces differences in retained-edge fraction and mean degree, but different edge weights and memberships remain, so it does not identify a causal effect of the composition operator.

Theorem 1 (Product edge localization). *Let $B = \beta + \gamma > 0$. If a product edge with threshold $0 < \theta < B$ is retained, then $d_{ij} < \sigma\sqrt{2\log(B/\theta)}$. For $\theta \geq B$ the strict retained set is empty; for $\theta \leq 0$ the threshold alone gives no finite cutoff.*

Proof. Since $a_i a_j \leq 1$ and $T_{ij}, C_{ij} \leq 1$, $w_{ij}^\times \leq B \exp[-d_{ij}^2/(2\sigma^2)]$. Combining $\theta < w_{ij}^\times$ with positivity and taking logarithms gives the bound; the endpoint cases follow from the same inequality.

This is an edge statement. A component-diameter claim additionally needs a bounded observed hop diameter; long transitive chains remain possible.

Corollary 1 (Conditional component bound). *In the finite product region, a connected component with at least two nodes, unweighted observed hop diameter $1 \leq h < \infty$, and geographic diameter D satisfies $D < hr_\theta^\times$, where $r_\theta^\times = \sigma\sqrt{2\log(B/\theta)}$. A singleton has $h = D = 0$ and is handled separately.*

For a non-singleton, the result follows from the triangle inequality along a path of at most h retained edges. It is a conditional post-hoc component bound, not an ex-ante compactness guarantee: the pipeline does not control h .

For the additive form, assume $\alpha > 0$. If $B < \theta < B + \alpha$, every retained edge satisfies

$$d_{ij} < r_\theta^+ = \sigma\sqrt{2\log\left(\frac{\alpha}{\theta - B}\right)}.$$

If $\theta \geq B + \alpha$, the strict retained set is empty; if $\theta \leq B$, no non-trivial threshold-induced geographic cutoff exists in general. Indeed, retention gives $\theta < \alpha G_{ij} + \beta T_{ij} + \gamma C_{ij} \leq \alpha G_{ij} + B$. Thus product composition has a finite geographic edge bound throughout its nonempty positive-threshold domain, whereas additive composition has such a bound only above the maximum non-geographic contribution. Neither statement alone implies better empirical clustering.

Require: Reports $R = \{r_i\}$; kernel scales $\sigma, \tau_t, \tau_F, \tau_E$;
weights β, γ (and α for the additive composition); quantile q ; neighbour
cap k ; Louvain/Leiden resolution ρ ; composition $m \in \{\text{product, additive}\}$

Ensure: Dispatch schedule S ; the algorithm below never reads incident
identity, benefit, deadline, or harm labels

- 1: $R_{\text{obs}} \leftarrow \emptyset; R_{\text{review}} \leftarrow \emptyset$
- 2: **for** each report $r_i \in R$ **do**
- 3: $a_i \leftarrow \mathbf{1}\{L_i, T_i \text{ observed}\}$ ▷ Section 2.1
- 4: **if** $a_i = 0$ **then**
- 5: $R_{\text{review}} \leftarrow R_{\text{review}} \cup \{r_i\}$ ▷ orange manual-review path
- 6: **else**
- 7: $R_{\text{obs}} \leftarrow R_{\text{obs}} \cup \{r_i\}$
- 8: **end if**
- 9: **end for**
- 10: Compute n_i^{corrob} from unique payloads in R_{obs} and set every Q_i by Eq. 1
- 11: $E_0 \leftarrow \text{BUILDSIMILARITYEDGES}(R_{\text{obs}}, m)$
using the fixed scales and weights above
BuildSimilarityEdges computes G, T, C on the full eligible pair matrix
- 12: $\theta \leftarrow q$ -quantile of the nonzero weights in E_0
- 13: $E_\theta \leftarrow \{(i, j, w_{ij}) \in E_0 : w_{ij} > \theta\}$
- 14: $E^\rightarrow \leftarrow \{(i, j, w_{ij}) \in E_\theta : j \text{ is among } i\text{'s top-}k$
neighbours in $E_\theta\}$
- 15: $E \leftarrow \{(i, j) : (i, j) \in E^\rightarrow \text{ or } (j, i) \in E^\rightarrow\}$
- 16: $\mathcal{G} \leftarrow (R_{\text{obs}}, E)$
- 17: $\mathcal{C} \leftarrow \text{Louvain/Leiden}(\mathcal{G}, \rho)$ ▷ predicted clusters, never incident truth
- 18: **for** each predicted cluster $C_k \in \mathcal{C}$ **do**
- 19: $P_k \leftarrow \text{PRIORITYSCORE}(C_k)$ ▷ Algorithm 2
- 20: **end for**
- 21: $S \leftarrow \text{Scheduler}(\{(C_k, P_k) : C_k \in \mathcal{C}\})$
scheduler sees predicted clusters and P_k only
- 22: **return** S

Algorithm 1: Fail-closed observable report pipeline (Fig. 1)

After S is fixed, the offline evaluator may join S and R_{review} with incident identity, benefit, deadline, and harm labels to compute ARI, NDCG@5, and dispatch outcomes. This dashed path is not available to the inference algorithm. In the current artifacts, the core schema requires location and event time, so the manual-review branch is a fail-closed design boundary rather than an empirically exercised missingness condition.

2.3 Duplicate-Aware Bounded Ranking

Exact duplicate payloads share a canonical fingerprint. Near duplicates are joined by deterministic connected components under the observable envelope; this transitive rule is not complete linkage. For a predicted cluster C_k , let $\mathcal{F}_k$

denote the resulting exact/near-duplicate evidence families inside that cluster. The revised estimator uses one confidence-gated priority per cluster and family maxima so that overlapping reports are not treated as disjoint people. Write $\tilde{N}_i = \min(N_i, N_{\text{ref}})$ and $\tilde{V}_i = \min(V_i, V_{\text{cap}})$ for the nonnegative clipped claims:

$$\begin{aligned}\bar{E}_k &= \frac{\sum_{g \in \mathcal{F}_k} \max_{i \in g} Q_i E_i}{\max(1, \sum_{g \in \mathcal{F}_k} \max_{i \in g} Q_i)}, & \bar{F}_k &= \max_{g \in \mathcal{F}_k} \max_{i \in g} Q_i F_i, \\ \bar{N}_k &= \min\left(1, \frac{\log(1 + \max_{g \in \mathcal{F}_k} \max_{i \in g} Q_i \tilde{N}_i)}{\log(1 + N_{\text{ref}})}\right), & \bar{V}_k &= \max_{g \in \mathcal{F}_k} \max_{i \in g} Q_i \tilde{V}_i, \\ P_k &= [1 + (\mu - 1) \tanh(\bar{V}_k/s)] (\omega_E \bar{E}_k + \omega_F \bar{F}_k + \omega_N \bar{N}_k).\end{aligned}\quad (4)$$

g indexes an evidence family and $i \in g$ a report in that family. $\bar{E}_k$ is a confidence-gated family aggregation, while $\bar{F}_k$, $\bar{N}_k$, and $\bar{V}_k$ are respectively maximum flood, normalized clipped population, and clipped vulnerability evidence. Here $N_{\text{ref}} > 0$ is the population reference, $V_{\text{cap}} \geq 0$ the vulnerability cap, $s > 0$ the saturation scale, $1 \leq \mu \leq 2$ the maximum amplification, and $\omega_E, \omega_F, \omega_N \geq 0$ sum to one. Therefore $\bar{E}_k, \bar{F}_k, \bar{N}_k \in [0, 1]$, $\bar{V}_k \in [0, V_{\text{cap}}]$, and $0 \leq P_k \leq \mu$. The candidate source snapshot associated with the RQ2/RQ3 artifacts specifies $\omega = (.34, .33, .33)$, $\mu = 2$, $s = 10$, $N_{\text{ref}} = 500$, and $V_{\text{cap}} = 50$. Maxima limit duplicate multiplicity, the logarithm compresses demand, clipping bounds individual evidence, and the hyperbolic tangent saturates vulnerability amplification. Reliability-aware truth discovery motivates Q_i but does not derive these choices [16]; they still require sensitivity and expert validation [13,6]. Comparators test raw multiplicity (legacy), single factors (urgency/population), an unbounded linear score, chance ordering, and travel time (nearest-first); none is an endorsed rescue policy.

- 1: **function** PRIORITYSCORE(cluster C_k)
- 2: $\mathcal{F}_k \leftarrow$ exact-fingerprint groups and deterministic near-duplicate connected components within C_k
- 3: Compute $\bar{E}_k, \bar{F}_k, \bar{N}_k, \bar{V}_k$ by Eq. 4
- 4: Compute P_k by Eq. 4
- 5: **return** P_k
- 6: **end function**

Algorithm 2: PRIORITYSCORE: duplicate-aware bounded priority (Eq. 4)

3 Experimental Design

Data contract and splits. The study uses two explicitly separated artifact suites. RQ1 uses the tracked generator-3.0 bundle, a semi-synthetic dataset with geographic anchors from the Copernicus EMSR848 Central Viet Nam activation [4]; Copernicus EMS, OpenStreetMap extracts, and the 2025 WorldPop constrained

population surface [17] provide geographic or historical anchors, while incidents, reports, duplicates, coordinated attacks, vulnerability, and labels are synthetic. It contains 80 runs with 16 latent incidents and 26,288 reports (17,901 genuine, 3,587 duplicate, and 4,800 coordinated nonincident reports), with 288–370 reports per run. Runs 1–20 are development, 21–40 calibration, and 41–80 the test split held out from calibration. These are within-generator tests, not mechanism-shift OOD transfer.

RQ2 and RQ3 use a separately generated Candidate-4.1 bundle. This bundle is a fully synthetic regional scenario defined by the generator and is not claimed to be EMSR848-anchored. It contains 80 datasets, 16 incidents per dataset, 30,229 reports in total, and 339–424 reports per dataset; the development, calibration, and test seeds are 1000–1019, 2000–2019, and 3000–3039. Across its splits there are 1,554 fake reports and 2,560 labeled duplicates (1,280 exact and 1,280 near). Each base dataset contains one five-report coordinated high-confidence campaign and one report for each of four low-confidence attack types; RQ2 stress scenarios perturb these inputs. For both suites, inference reads only observable `algorithm_input.json`; ground truth is opened only after predictions are fixed.

Fixed clustering stresses. The 40 held-out generator-3.0 test runs are also evaluated under seven fixed conditions: the unmodified control; independent Gaussian location noise with 100 m or 300 m standard deviation on each local axis; independent Gaussian timestamp noise with 15 min or 60 min standard deviation; and exact transport copies at total multiplicity two or five, where only the observable report identifier changes. Five configurations selected before this stress evaluation are retained without tuning: Product Louvain, Additive Louvain, matched-density Additive, Product Leiden, and geo-time DBSCAN. Each method receives the same deterministic perturbation for a run and condition, yielding $40 \times 7 \times 5 = 1,400$ fits. These controlled levels characterize the tested pipeline and are not estimates of field measurement error or duplicate prevalence.

Selection and comparators. The current notebook evaluates complete small grids on the 20 calibration runs and selects mean ARI, breaking ties by standard deviation and canonical configuration JSON. Product and additive use fixed $\tau_F = .25$, $\tau_E = .35$, and $\beta = \gamma = .5$. The Product grid varies $\sigma \in \{700, 1200\}$ m, $\tau_t \in \{30, 60\}$ min, $q \in \{.90, .95\}$, and $k \in \{8, 16\}$; the Additive grid varies $\alpha \in \{1/3, 1\}$, the same σ, q, k sets, and fixes $\tau_t = 60$ min. Both fix resolution at 1.2. Selected product Louvain uses $\sigma = 700$ m, $\tau_t = 60$ min, quantile .90, $k = 8$, and resolution 1.2. Selected additive Louvain uses $\alpha = 1$, $\sigma = 700$ m, $\tau_t = 60$ min, quantile .95, $k = 8$, and resolution 1.2. Because selected quantiles differ, this is an independently calibrated pipeline comparison, not a matched-density operator comparison. Comparators are convex similarity, Product Leiden, geo-time DBSCAN, DBSCAN/HDBSCAN on scaled observable features, spatial agglomeration, and coordinate K-means. Geo-time DBSCAN lacks ST-DBSCAN’s non-spatial criterion. Together these variants separate composition

Table 1. Artifact construction. Geographic anchors do not make simulated reports or outcomes into field observations.

Layer	Generator 3.0 (RQ1)	Candidate 4.1 (RQ2/RQ3)
Geographic context	EMSR848 context, OSM anchors, WorldPop cells	Synthetic geometry; no EMSR848 claim
Reports/linkage	Synthetic incidents, reports, duplicates, campaigns, labels	Synthetic incidents, reports, duplicates, attacks
Operational truth	Simulated need and vulnerability; population is not trapped-person truth	Simulated benefit, deadlines, service, access, harm
External validity	Geographic plausibility only; no source snapshot or row linkage	Controlled stresses; no field sample or transfer

and community detection, reduce the Product/Additive density imbalance, and supply density-, hierarchy-, and geometry-based controls.

Endpoints and inference. ARI on incident-linked reports [7] is the primary endpoint; NMI, pairwise F1, geographic diameter, runtime, false destinations, fake absorption, noise rejection, retained-edge fraction, and mean degree are descriptive or operational diagnostics. RQ1 paired differences use 10,000 bootstrap resamples, Wilcoxon tests, and Holm correction; RQ2/RQ3 use 5,000 resamples. Reported p-values follow exported families. RQ1 adjusts all nine Product-Louvain contrasts, with the product/additive/Leiden contrasts primary. RQ2 adjusts three alignment endpoints within comparator and five robustness endpoints within scenario; false-priority lift is descriptive. RQ3 adjusts 14 dispatch endpoints within each candidate/comparator/context block and aggregates resource conditions within seed, retaining 40 pairs; the original 120 seed-condition analysis is diagnostic. RQ2 uses oracle groups to isolate ranking, while RQ3 uses predicted clusters and independent simulated outcomes.

Fixed-stress ARI and pairwise F1 use original incident-linked reports so copy multiplicities share an evaluation unit; all-payload and graph diagnostics are descriptive. Each condition is paired with its method’s control over 40 runs. The 5,000-resample unadjusted CIs describe within-method change, not between-method superiority.

4 Results

4.1 RQ1: Clustering Accuracy and Fixed Stresses

Table 2 reports the held-out clustering summary.

Additive Louvain has the highest mean ARI. Product minus independently tuned Additive Louvain is -0.01184 with an unadjusted bootstrap CI $[-0.02257, -0.00240]$, with 9 wins, 19 ties, and 12 losses. The Wilcoxon test

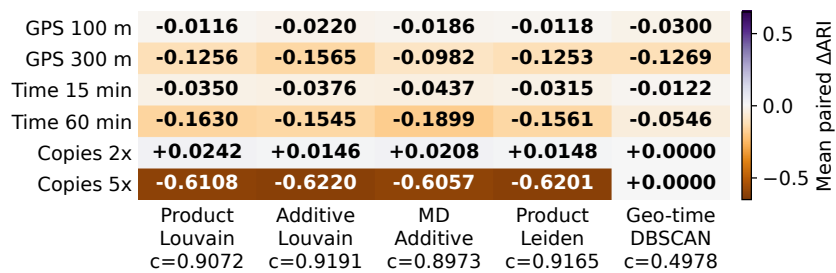
Table 2. RQ1 benchmark-test summary over 40 runs. Intervals are bootstrap 95% confidence intervals; diameter is descriptive.

Method	ARI [95% CI]	Pair F1	Mean diam. (km)
Additive Louvain	.9191 [.8995,.9381]	.9247	.3477
Product Leiden	.9165 [.8979,.9351]	.9223	.3547
Convex Louvain	.9135 [.8940,.9320]	.9195	.3547
Product Louvain	.9072 [.8864,.9271]	.9138	.3585
Matched-density Additive	.8973 [.8763,.9175]	.9046	.3743
Coordinate K-means	.8176 [.7976,.8384]	.8292	.2977
Spatial agglomerative	.6235 [.5964,.6494]	.6506	4.1648
HDBSCAN	.5205 [.4565,.5811]	.5787	1.2527
Geo-time DBSCAN	.4978 [.4589,.5366]	.5450	.6244
DBSCAN, all features	.4741 [.4273,.5201]	.5488	.9340

has raw $p = .0987$ and Holm-adjusted $p = .2560$; the interval and test use different procedures, and the adjusted test does not establish a composition advantage. Product minus matched-density Additive is $+0.00989$ with unadjusted CI $[-0.00100, 0.02082]$ and Holm-adjusted $p = .2560$; Product Leiden versus Product Louvain has Holm-adjusted $p = .1120$. Product and matched-density Additive have similar retained-edge fractions (.03011 and .03021) and mean degrees (9.90 and 9.94), making their contrast a density diagnostic rather than a causal estimate of the composition operator.

The product pipeline has zero fake absorption and zero fake rejection but a 16.49% false-destination rate; fake reports avoid genuine destinations while still forming fake-only destinations. This is an adverse operational result, not evidence of misinformation robustness.

Figure 2 reports the fixed clustering stresses as paired changes from each method’s control. All tested location and timestamp noise levels reduce mean ARI, and no method has the highest mean ARI in every condition.

**Fig. 2.** Mean paired change in RQ1 ARI relative to each method’s control over 40 held-out runs. Cells use original incident-linked reports; negative values indicate degradation; labels give control ARI, and zero change does not imply high absolute accuracy. MD Additive denotes matched-density Additive. Intervals are not shown and no between-method test is implied.

Twofold copies slightly increase ARI for all four graph variants, whereas fivefold copies reduce their means to approximately .30; geo-time DBSCAN remains at its lower control mean. These two tested multiplicities show a non-monotone response, not a beneficial duplication effect or an identified collapse threshold. For Product Louvain, the fivefold change is $-.6108$ with an unadjusted paired-bootstrap 95% CI $[-.6318, -.5881]$. Thresholding and endpoint-wise top- k sparsification are possible explanations, but the protocol does not isolate a mechanism. An independent, unpooled rerun changes only the aggregate twofold-copy graph values slightly; the GPS, timestamp, fivefold-copy, and cross-condition conclusions remain unchanged. Thus score-level invariance does not imply clustering invariance or end-to-end robustness.

4.2 RQ2: Copy Invariance and Benefit Alignment

Table 3. RQ2 robustness means over the held-out Candidate-4.1 test split. n counts target-perturbation rows; ΔP is absolute normalized priority drift, Δr is normalized mean-rank drift, and churn is top-five churn. R/L denote revised/legacy; lower is better.

Scenario	n	ΔP_R	ΔP_L	Δr_R	Δr_L	Churn _R	Churn _L
Exact duplicate 2×	40	.0000	.0026	.0000	.0015	.0000	.0000
Exact duplicate 5×	40	.0000	.0086	.0000	.0046	.0000	.0100
Exact duplicate 10×	40	.0000	.0148	.0000	.0075	.0000	.0150
Near duplicate	320	.0016	.0055	.0004	.0017	.0006	.0019
Contradictory F/E	40	.0484	.0611	.0171	.0290	.0450	.1100
Missingness/zero-imputation	628	.0063	.0075	.0019	.0025	.0029	.0064
Low-confidence E inflation	40	.0034	.0012	.0008	.0006	.0050	.0000
Low-confidence F inflation	40	.0026	.0072	.0000	.0040	.0000	.0150
Low-confidence N inflation	40	.0696	.0140	.0240	.0071	.0600	.0150
Low-confidence V inflation	40	.0758	.0074	.0263	.0040	.0500	.0150
Coordinated high-confidence	40	.1688	.0309	.0175	.0006	.0350	.0000

Under oracle incident grouping, revised NDCG@5 is .6669 (CI [.6306,.7018]), Spearman is -.0304 (CI [-.1165,.0549]), and top-five overlap is .285 (CI [.2349,.3400]). Revised exceeds legacy on NDCG@5 by .0121, but the Holm-adjusted contrast is not significant. Its NDCG@5 point estimate is lower than population-only (.6709) and random (.6763); these point estimates do not by themselves establish significant differences. Because oracle grouping removes upstream clustering errors from this RQ, the weak alignment cannot be attributed solely to clustering.

Exact duplication at 2×, 5×, and 10× produces zero revised priority drift when evidence grouping and inputs are fixed. A coordinated high-confidence campaign instead produces drift .1688 and false-priority lift .8828, increasing revised drift by .1379 relative to legacy (Holm-adjusted $p \approx 9.1 \times 10^{-12}$). Bound-ness and exact-copy invariance therefore do not guarantee ranking reliability.

Table 3 expands the robustness audit beyond the two headline cases. The revised score is invariant to exact-copy multiplicity and usually changes less under near duplication, missingness, contradictory evidence, and low-confidence flood inflation. Its bounded form is nevertheless more sensitive than the legacy score to low-confidence urgency, population, or vulnerability inflation and to coordinated high-confidence reports. These are controlled synthetic perturbations, not estimates of attack prevalence or field error.

Together, the oracle-group alignment and perturbation results separate narrow invariance from empirical utility: exact-copy invariance holds under fixed inputs, while benefit alignment remains unestablished.

4.3 RQ3: Dispatch Outcomes

After averaging the three resource scenarios within each seed, revised priority on the Product partition increases simulated harm by 14.34 relative to legacy while reducing the deadline-miss rate by 1.88 percentage points; neither contrast is significant after Holm correction ($p = .563$ and $p = .247$, respectively). Revised is worse than nearest-first on both harm (Holm-adjusted $p = 4.1 \times 10^{-5}$) and deadline misses (Holm-adjusted $p = 1.3 \times 10^{-6}$). Product is also worse than matched-density Additive under revised priority on harm (Holm-adjusted $p = .000168$), and worse than oracle on the corresponding outcomes. Predicted split, merge, and fake-only destinations therefore have measurable scheduling consequences.

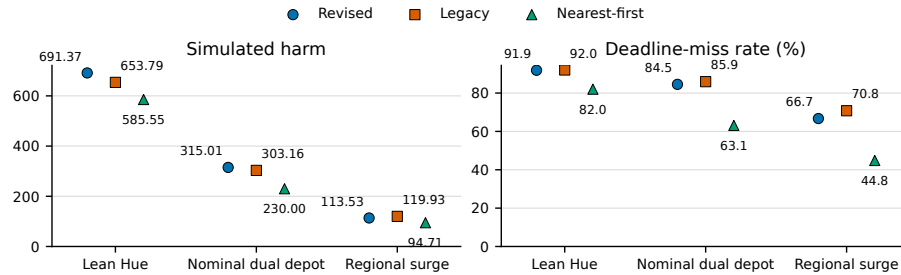


Fig. 3. Descriptive Product-partition means over 40 test seeds for each resource condition. Points are not additional independent samples or between-policy tests. Primary inference averages the three conditions within each seed and uses 40 seed-level pairs. Lower is better for both panels.

Figure 3 shows that nearest-first has lower mean harm and deadline-miss rate in every tested condition. Revised priority lowers mean deadline misses relative to legacy in the nominal and surge conditions and lowers mean harm only in surge. One plausible explanation is that need-based ranking does not fully represent

travel costs and scheduling constraints, but this experiment neither isolates that mechanism nor establishes nearest-first as a field policy.

4.4 Sensitivity of Heuristic Constants

To assess robustness to the fixed constants in Q_i (Section 2.1) and P (Eq. 4), we define a local one-at-a-time sensitivity analysis on the Candidate-4.1 ranking task. Each valid constant is perturbed around its reference value while all other constants are held fixed, and NDCG@5 is recomputed on the same 40 test seeds. The ω variants perturb one component at a time and renormalize the vector; μ is restricted to its policy-valid interval $[1, 2]$. The locked rerun evaluates 21 configurations on the same 40 held-out seeds, yielding 840 seed-level observations. Table 4 reports the tested levels and mean-NDCG@5 ranges; no value is inferred from unrelated fixed-configuration comparisons.

Table 4. Candidate-4.1 one-at-a-time sensitivity of mean NDCG@5 to fixed heuristic constants. Each range is the minimum–maximum mean over the tested levels, including the reference level, on 40 test seeds. All spans are below 0.01 under the locked rerun.

Constant	Reference/default value	Tested levels	Metric range
b_0 (Q_i intercept)	−0.20	[−0.24, −0.16]	[.6653, .6672]
b_1 (image weight)	1.40	[1.12, 1.68]	[.6669, .6687]
b_2 (corroboration weight)	0.90	[0.72, 1.08]	[.6667, .6679]
ω_E	.34	component $\times \{.8, 1, 1.2\}$, renormalized	[.6663, .6714]
ω_F	.33	component $\times \{.8, 1, 1.2\}$, renormalized	[.6625, .6679]
ω_N	.33	component $\times \{.8, 1, 1.2\}$, renormalized	[.6663, .6697]
μ (vulnerability amplification)	2	$\{1.6, 1.8, 2.0\}$	[.6644, .6683]
s (vulnerability saturation)	10	[8, 12]	[.6665, .6713]
N_{ref}	500	[400, 600]	[.6669, .6669]
V_{cap}	50	[40, 60]	[.6669, .6669]

All mean-NDCG@5 spans are below .01 over the tested levels; the largest is .00546 for ω_F . The unchanged reported mean for N_{ref} and V_{cap} does not imply invariant per-seed scores or rankings. More generally, local numerical stability coexists with weak benefit alignment and does not validate the scoring rule or the constants outside these levels.

5 Discussion and Threats to Validity

The safeguards have stage-specific scope. Product gating gives a finite conditional edge bound, yet no composition advantage is established and the fixed stresses expose clustering sensitivity. The scheduler exposes consequences of both imperfect destinations and an unvalidated ranking objective. Clustering accuracy, score boundedness, and local parameter stability answer distinct questions and cannot substitute for downstream validation.

All data and outcomes are synthetic, and the separate RQ1 and RQ2/RQ3 generators do not test OOD transfer or field performance. RQ1 lacks an empirical

missingness test; the score lacks expert validation; and extraction, deployment cost, source authenticity, and actual harm remain unmeasured. Geographic anchors add context, but no raw snapshots, checksums, or row lineage. Table 1 therefore documents construction, not provenance or representativeness. These limits preclude policy or deployment claims.

6 Conclusion

Product gating gives conditional edge-localization bounds, and family-wise scoring is copy-invariant under fixed grouping and inputs. Neither ensures clustering stability, benefit-aligned ranking, or better dispatch: fivefold copies degrade graph clustering, ranking remains weak under oracle grouping, and nearest-first has lower harm and deadline misses in every tested resource condition. These endpoints require separate evaluation; clustering accuracy and score boundedness are not downstream validation. Applicability remains limited to the tested synthetic generators and configurations, without establishing superiority, policy validity, or field readiness.

Data and Code Availability

The Zenodo artifact [9] corresponds to GitHub Release v1.0.1. The RQ1 benchmark snapshot is `6ac75c2`; the generator and external benchmark are at `a6be3e9`. The public `clean` commit `5fad10d` adds primary fixed-stress results and an independent, unpooled rerun. The primary notebook is included; the rerun notebook is pending. These additions are outside `v1.0.1`; raw anchors are not redistributed. Code is MIT licensed; the contribution follows the Springer agreement, and other content retains its stated terms.

Acknowledgments. This study is funded by the Can Tho University, Code: THS2026-68.

Disclosure of Interests. The authors have no competing interests to declare.

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